

CloudWalk Payments Inc.

AML Program Update — High-Risk MCCs, Cross-Border E-Commerce, Device Sharing & Chargebacks

Prepared for: BSA/AML Compliance Committee and Sponsor Bank (Lead Bank) | Classification: Internal — Confidential | Date: November 2025

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Scope: This document proposes targeted updates to CloudWalk's BSA/AML Program addressing high-risk MCC controls, cross-border e-commerce rules, device-sharing detection, chargeback governance, customer risk rating methodology, recordkeeping, USA PATRIOT Act Sections 314(a)/314(b), prepaid access controls, and the sponsor-bank reporting chain. All thresholds below are proposed calibrations based on the confirmed transaction dataset (32,348 transactions, 501 customers, 100 merchants, 630 cards) and are subject to Compliance Committee and Lead Bank approval prior to production enforcement.

1. Controls by High-Risk MCC

The dataset confirms six MCCs flagged `high_risk_mcc_flag=True`. Each requires MCC-specific velocity limits, alert thresholds, documentation, and — where the merchant operates as a money services business — verified MSB registration.

4829 — Money Transfer (2,924 txns)

Merchants: RemitFast #325, #326, #331, #343, #359, #361, #363, #382, #390

- Velocity limit: transactions $\geq$ \$3,000 trigger mandatory Travel Rule data collection (31 CFR 1010.410 — originator and beneficiary name, address, account number) before the transfer is released.
- Velocity limit: aggregate customer volume $>$ \$10,000 within a rolling 24-hour window requires supervisory review before further transactions are approved.
- Alert threshold: $\geq$ 3 transactions in the \$980–\$995 range from a single customer within 7 days (structuring pocket — see Section 5 SQL Query 3).
- Alert threshold: $\geq$ 3 distinct `customer_ids` on the same `device_id` transacting at MCC 4829 within 60 minutes (mule aggregation — see Query 2).
- Documentation: valid `msb_registration_number` verified at onboarding and at every 12-month KYB refresh; suspend processing for any MCC 4829 merchant lacking registration within 30 days of detection.
- Documentation: source-of-funds attestation required for any single customer transfer exceeding \$3,000.

6051 — Quasi Cash / Crypto (1,687 txns)

Merchants: CryptoBuy24 #304, #315, #332, #353, #381

- Velocity limit: \$2,000 per customer per day; \$10,000 per customer per rolling 30 days — amounts above trigger EDD referral, not automatic block.
- Alert threshold: ≥ 3 transactions in the \$980–\$995 range within 7 days from a single customer (structuring pocket).
- Alert threshold: any single transaction $> \$5,000$, or any cross-border MCC 6051 transaction (e.g., CN corridor present in dataset).
- Documentation: enhanced source-of-funds narrative required for any transaction $> \$1,000$; independent UBO verification and OFAC screening for the merchant at onboarding and 12-month refresh.
- Instant funding / OCT into MCC 6051 remains blocked pending completion of the formal Product Risk Assessment (Section 5 of the case).

7995 — Gambling / Casino (1,987 txns)

Merchants: LuckyAce Casino #319, #333, #334, #336, #378, #387

- Alert threshold: $\geq \$5,000$ in same-customer activity within 24 hours.
- Alert threshold: rapid deposit-then-withdrawal cycles within 24 hours with $< 10\%$ net play (classic layering-through-gaming pattern).
- Alert threshold: ≥ 3 transactions $> \$3,000$ from a single customer within 24 hours.
- Documentation: valid state gaming license and AML program attestation verified at onboarding and annual refresh, per FinCEN guidance for money services at casinos and card clubs (31 CFR 1021).

5967 — Direct Marketing / Teleservices (2,880 txns)

Merchants: DirectTel Services #314, #316, #322, #323, #329, #344, #377, #379, #394

- Alert threshold: recurring/subscription billing exceeding \$500/month per customer without a clearly documented product or service.
- Alert threshold: chargeback ratio for the merchant exceeding 3% over a rolling 30-day window (this MCC has historically elevated dispute rates).
- Alert threshold: ≥ 5 declined-then-approved authorization attempts within 30 minutes on the same card (overlap with card-testing typology, Query 1).
- Documentation: verification of legitimate business model, telemarketing registration where applicable, and clear product/service descriptors at underwriting and 12-month refresh.

6011 — ATM / Financial (2,582 txns)

Merchants: ATM Express #302, #320, #346, #370, #373, #384, #392, #396

- Alert threshold: single-customer cash withdrawals exceeding \$1,000/day.
- Alert threshold (layered — not standalone): ATM withdrawal followed within 72 hours by a remittance to MCC 4829 equal to $\geq 80\%$ of the withdrawn amount. The MCC 6011 + 4829 pairing alone is not discriminating (489 of 501 customers match it) and must never be used as a standalone alert; it requires the time-window and ratio criteria above (Query 7).
- Documentation: msb_registration_number mandatory for all non-bank ATM operators, verified at onboarding and 12-month refresh; two merchants (M3020-AE, M3070-BR) are currently unregistered and must be remediated or suspended within 30 days.

5944 — Jewelry (1,242 txns)

Merchants: Empire Jewelry #305, #306, #355, #389

- Alert threshold: single transaction $\geq \$10,000$ (dealer-in-jewels reporting nexus under the BSA, 31 CFR 1027, distinct from the CTR cash-reporting requirement).
- Alert threshold: ≥ 3 transactions just under \$10,000 from a single customer within 7 days (structuring); cross-border high-value shipments.
- Documentation: dealer/precious-metals registration verification where applicable; enhanced documentation (invoice, appraisal, source-of-funds) required for transactions $> \$5,000$.

2. Cross-Border E-Commerce

Rules are tiered by country according to current FATF and OFAC status, cross-referenced against the countries actually present in the transaction dataset.

Country	FATF / OFAC Classification	Action	Rule
IR, KP, SY, CU	OFAC comprehensive sanctions; IR and KP also on FATF Black List	BLOCK	Real-time authorization decline. Screened against OFAC SDN and comprehensive country lists at the acquirer/network level. No customer risk rating can override a block.
VE	FATF Grey List (Oct	EDD MANDATORY	EDD required before

Country	FATF / OFAC Classification	Action	Rule
	2025)		any transaction above \$500 is approved for a VE-domiciled customer or a merchant with VE beneficial ownership; ongoing enhanced monitoring; risk_rating upgraded to high, kyc_level to enhanced.
RU	OFAC targeted / sectoral sanctions	EDD MANDATORY	EDD required at onboarding for any RU-nexus customer, merchant, or UBO; enhanced screening against OFAC sectoral sanctions identifications (SSI) list; senior compliance approval to continue relationship.
AE, TR	Elevated ML/TF-risk transit jurisdictions; not currently FATF-listed	EDD REQUIRED	EDD required at onboarding when monthly cross-border volume with AE/TR counterparties exceeds \$2,000; ongoing transaction monitoring at a lower alert threshold than standard-monitor countries.
CA, BR, GB, CN, MX	Standard cross-border partner / trading jurisdictions	MONITOR	Standard cross-border transaction monitoring; alert if a customer's cross-border volume with these countries exceeds 50% of total

Country	FATF / OFAC Classification	Action	Rule
			account activity, or if MCC risk-stacks with high-risk MCCs (e.g., CN + MCC 6051).

When to block vs. alert vs. EDD

- Block: real-time authorization decline at the point of sale/checkout. Applied automatically to OFAC comprehensive-sanctions countries (IR, KP, SY, CU) regardless of the customer's existing risk rating or transaction history — no override is permitted without a documented OFAC license.
- EDD: triggered at onboarding, or immediately upon detection of a nexus to an EDD-mandatory or EDD-required country (VE, RU, AE, TR). The transaction is not blocked, but processing continues only under enhanced monitoring, senior compliance approval, and a periodic 90-day case review.
- Alert: a transaction-level alert is generated for investigator review within the applicable SLA tier (P1/P2/P3, see Task 5.1). The transaction is not blocked pending review, but must receive a documented disposition (close / escalate / SAR) within the SLA window.

3. Device Sharing

Device-sharing thresholds are calibrated against the dataset's confirmed extreme case: 12 distinct customers transacting on a single device_id at MCC 4829, well in excess of any reasonable shared-household or shared-computer explanation.

- Customer-per-device alert threshold: ≥ 3 distinct customer_ids on the same device_id within a 60-minute window generates a mule-aggregation alert for investigator review.
- Customer-per-device hard-block threshold: ≥ 5 distinct customer_ids on the same device_id within 60 minutes triggers an automatic hold on further transactions from that device pending manual investigation. The dataset's 12-customer case exceeds both thresholds by a wide margin and should have triggered an automatic hold under this rule.
- Action when device_id appears in MCC 4829 or 6051 specifically: any device with ≥ 2 distinct customers transacting in either MCC within a rolling 24-hour window triggers EDD review of all associated customer_ids; at ≥ 3 distinct customers, new transactions from that device are frozen pending investigation, and the finding is escalated to AML Transaction Monitoring / Fraud.

4. Chargebacks

Thresholds are calibrated against the dataset's confirmed baseline chargeback ratio of ~0.73% (236 of 32,348 transactions) and the confirmed outlier merchant (M3030 — ElectroHub #330) at a 12% ratio.

- Yellow alert: merchant chargeback ratio exceeds 3% (double the baseline) over a rolling 30-day window — triggers a documented merchant review.
- Red alert / escalation: merchant chargeback ratio exceeds 5% over the rolling 30-day window — triggers escalation to Head of Merchant Underwriting and a formal remediation plan request from the merchant.
- Automatic suspension: merchant chargeback ratio $\geq 8\%$ sustained across two consecutive rolling 30-day windows, or $\geq 10\%$ in any single rolling 30-day window, triggers immediate processing suspension pending investigation. The dataset's confirmed 12% outlier meets this immediate-suspension criterion.
- The rolling 30-day window is recalculated daily. Reinstatement after suspension requires a root-cause remediation plan approved by the Head of Merchant Underwriting and, where applicable, confirmation from the sponsor bank.

5. Customer Risk Rating

Risk rating is calculated using a weighted point methodology, then mapped to a risk band. Any single “mandatory” trigger overrides the computed score and forces a high rating regardless of total points.

Risk Factor	Condition	Points
PEP status	pep_flag = True	+40 (mandatory high rating)
Sanctions match score	0.3 – 0.5	+15
Sanctions match score	0.5 – 0.7	+30
Sanctions match score	> 0.7	+50 (mandatory high + immediate EDD)
Country risk	OFAC comprehensive (IR, KP, SY, CU)	+50 (mandatory high)
Country risk	FATF Grey (VE, updated per current list)	+35 (mandatory high)
Country risk	Elevated (AE, TR, RU)	+20
Country risk	Standard / no listing	+0
Doc issue country	Matches OFAC/FATF list above (stacks with country field)	+30
MCC usage	High-risk MCC is customer's	+15

Risk Factor	Condition	Points
Volume deviation	primary spend category Actual volume > 150% of expected_monthly_volume_u sd	+15
Volume deviation	Actual volume > 300% of expected_monthly_volume_u sd	+25
Card product	Prepaid card	+10

Score bands

Score Band	Risk Rating	Notes
0 – 20	Low	24-month refresh cadence; standard monitoring.
21 – 50	Medium	12-month refresh cadence; standard-to-enhanced monitoring depending on trend.
> 50, or any single mandatory trigger	High	6-month refresh cadence (quarterly if PEP or sanctions_match_score > 0.7); mandatory EDD and senior approval.

Periodic refresh cadence by tier

Risk Tier	Refresh Cadence	Additional Requirement
Low	24 months	Automated re-screening against sanctions/PEP lists monthly regardless of refresh date.
Medium	12 months	Automated re-screening monthly; transaction-pattern review at refresh.
High	6 months	Automated re-screening weekly; senior compliance sign-off required to continue relationship.
PEP or sanctions_match_score > 0.7	Quarterly (3 months)	Continuous transaction monitoring; immediate escalation on any new adverse-media or sanctions hit.

Mandatory upgrade triggers (override the computed score)

- Venezuela (VE) or Iran (IR) nexus, via either the country field or doc_issue_country field: immediate upgrade to risk_rating=high and kyc_level=enhanced, regardless of the computed point total.
- sanctions_match_score > 0.7: immediate upgrade to risk_rating=high with EDD initiated within 24 hours of detection.
- expected_monthly_volume_usd > \$10,000 while kyc_level ≠ enhanced: automatically flagged for mandatory KYC-level upgrade within 30 days of detection.

6. Recordkeeping

Mandatory fields to retain

- Transaction fields: txn_id, txn_timestamp_utc, card_id, customer_id, merchant_id, mcc, amount_usd, merchant_country, acquirer_country, channel, pos_entry_mode, device_id, ip_address, response_code, status, is_chargeback, is_refund.
- Customer (CIP) fields: customer_id, full_name, dob, ssn_last4, country, document_type, doc_issue_country, doc_expiry_date, kyc_level, risk_rating, pep_flag, sanctions_match_score, last_kyc_refresh.
- Merchant (KYB) fields: merchant_id, dba_name, primary_mcc, incorporation_country, beneficial_owners_json, ofac_match_score, msb_registration_number, high_risk_mcc_flag, last_kyb_refresh.

Retention periods (BSA-aligned)

- SARs, CTRs, and all supporting documentation: 5 years from the date of filing (31 CFR 1020.320 / 1010.430).
- CIP records: 5 years after the date the account is closed.
- Travel Rule records (31 CFR 1010.410): originator name, address, and account number, plus beneficiary name, address, and account number, retained for 5 years for all MCC 4829 transfers ≥ \$3,000, and made available to the next financial institution in the payment chain upon request.

7. USA PATRIOT Act Sections 314(a) and 314(b)

Section 314(a) — government information request

- Upon receipt of a FinCEN 314(a) request, the request is routed immediately to the CCO / AML Program Manager.

- The customer and transaction database is searched against all names/identifiers in the request within 24 hours of receipt.
- Any positive match is reported to FinCEN within the mandatory 14-business-day deadline from the date of the request.
- No match, no contact: the subject of a 314(a) request is never informed, directly or indirectly, that a request was received.
- The search methodology and results (including negative results) are logged and retained for 5 years.

Section 314(b) — voluntary inter-institutional information sharing

- CloudWalk (or the Lead Bank on its behalf) maintains an active annual FinCEN 314(b) registration, a prerequisite for the safe harbor.
- 314(b) exchanges are limited to matters involving suspected money laundering or terrorist financing activity — not general due diligence requests.
- Every proposed 314(b) exchange requires CCO approval before information is shared, and the fact, date, and counterparty of the exchange (not the underlying customer data) is logged in a 314(b) registry for audit purposes.

8. Prepaid Access Controls (31 CFR 1010)

CloudWalk has 227 prepaid cards in issue; 40.8% of prepaid transaction volume already occurs in high-risk MCCs, which compounds the inherent anonymity/portability risk of the prepaid product.

- Per-customer spend tracking: \$1,000 daily limit and \$10,000 monthly limit, enforced with an automatic hold once either threshold is reached.
- Customer identity thresholds: full CIP (name, DOB, address, government ID) is required before any prepaid program allows access exceeding \$1,000 in a single day or \$10,000 in a rolling 12-month period, consistent with the Prepaid Access Rule's non-anonymous-access thresholds (31 CFR 1010.100(ff)(4)). Below those thresholds, access is capped at \$200 per transaction and \$1,000 per month without full CIP.
- Reload velocity monitoring: maximum 3 reloads per day and \$2,000 in reloads per day per card; an alert is generated whenever a reload is immediately (within 1 hour) followed by spend at a high-risk MCC.

9. Sponsor Bank Reporting Chain

CloudWalk operates under a program-manager / sponsor-bank model in which Lead Bank (FDIC member) is the covered financial institution of record for SAR filing purposes.

- Internal detection: once an alert is confirmed as suspicious by CloudWalk’s AML investigation team, it is escalated to the Lead Bank BSA Officer within 48 hours of confirmation, documented in the shared case management system.
- SAR filing: Lead Bank reviews the escalated case and files the SAR with FinCEN within the 30-calendar-day window measured from the date of initial detection of facts that may constitute a basis for filing (31 CFR 1020.320); continuing SARs are reviewed every 90 days for as long as the suspicious activity continues.
- Governance: CloudWalk and Lead Bank hold a joint quarterly governance committee meeting to review filing volumes, 48-hour escalation SLA adherence, false-positive/false-negative trends, and any regulatory correspondence.
- This chain, including the 48-hour internal SLA and Lead Bank’s independent SAR-filing authority, is documented formally in the BSA/AML Program manual and in the CloudWalk–Lead Bank Program Management Agreement.

10. Root Cause: Data Pipeline Timestamp Integrity

10.1 Finding

A population-level scan of the Merchants_KYB table (100 confirmed merchants) identifies **30 of 100 merchants (30.0%)** with `last_kyb_refresh` dated *before* `onboarding_date` — a logical impossibility under any correct KYB workflow, since a merchant cannot receive a periodic KYB refresh prior to being onboarded. Deltas range from 4 to 76 days (median $\approx$ 30 days), across 26 distinct date pairs. This is not a data-entry outlier confined to one or two records; it is a systemic pattern affecting nearly a third of the merchant population and warrants remediation at the pipeline level, not case-by-case correction.

Two of the affected merchants — **M3004** (delta: 64 days) and **M3056** (delta: 59 days) — intersect with prior OFAC/UBO critical-case analysis in this case file. Their KYB-refresh evidentiary basis should be treated as provisionally unreliable pending the correction workflow in Section 10.3, and any conclusions drawn from `last_kyb_refresh` in those case narratives should be re-verified once corrected values are available.

10.2 Pattern Analysis

The following observations bear on causation and rule out (or support) specific hypotheses:

- **No single repeated constant.** The 30 anomalous records carry 26 distinct `last_kyb_refresh` values (range: 2025-09-01 to 2025-10-29). This rules out a simple null-substitution or single hardcoded default (e.g., 1900-01-01 or a fixed sentinel date) as the sole mechanism.
- **Portfolio-wide refresh dates are bounded to a narrow 61-day window, independent of onboarding.** Across *all* 100 merchants — anomalous and normal alike — `last_kyb_refresh` falls entirely within **2025-09-01 to 2025-10-31**. `onboarding_date`, by contrast, spans **2025-03-10 to 2025-11-25** (a $\sim$ 261-day range). A refresh field that is

supposed to track individual onboarding-relative events should not be uniformly compressed into a 61-day band while onboarding is not. This is the single strongest signal in the dataset.

- **Anomalies concentrate almost entirely in the newest onboarding cohort.** 100% of merchants onboarded in November 2025 (13/13) and 79% of merchants onboarded in October 2025 (15/19) fall into the anomalous group, versus 0% of merchants onboarded March–August 2025. Merchants onboarded *after* the refresh window closed (October 31) mathematically cannot have a valid post-onboarding refresh timestamp if the refresh field was populated by a process bounded to that window.
- **Day-level clustering within the anomalies.** Several anomalous merchants share an identical last_kyb_refresh value (e.g., three merchants stamped 2025-09-01; three stamped 2025-09-06; pairs on 2025-09-09, 2025-10-18, 2025-10-20, and 2025-10-22). This cohort-style clustering is more consistent with a scheduled batch process touching multiple merchant records on the same run date than with an individually-triggered, per-merchant event.
- **No material concentration by risk tier or MCC.** High-risk-rated merchants make up 47% of the anomalous group versus 41% of the full portfolio — a modest, not material, skew. Incorporation-country and high-risk-MCC-flag distributions are similarly proportionate to the overall population. This indicates the anomaly is a pipeline/timing artifact rather than a pattern targeted at, or caused by, any specific risk segment.

10.3 Root-Cause Hypotheses (Ordered by Plausibility)

No single cause is asserted without corroborating system evidence (service logs, job schedules, or ETL run history), which CloudWalk’s Data Engineering team should provide to adjudicate between the hypotheses below. In order of support from the data:

- **Asynchronous/batch KYB-refresh job decoupled from onboarding completion (most plausible).** A scheduled refresh process appears to have executed against the merchant portfolio within a bounded operational window (approximately September 1 – October 31, 2025), stamping last_kyb_refresh based on the *job’s run date* rather than a value computed relative to each merchant’s individual onboarding event. Merchants onboarded during or after this window inherit a refresh timestamp generated before their own onboarding was complete. The bounded 61-day window applying uniformly across the entire portfolio, combined with same-day clustering among several anomalous records, is the strongest evidence for this hypothesis.
- **Database default value or backfill/seeding artifact.** During an initial data load, migration, or backfill exercise, last_kyb_refresh may have been populated using a fixed “as-of” extraction or go-live window rather than being derived per-merchant from actual refresh events. This would produce a similar bounded-window signature. It is ranked second rather than first because the 26 distinct dates and day-level cohort variation suggest a computed or randomized process rather than one universal static default — though a randomized backfill routine cannot be excluded and would look identical to Hypothesis 1 from the data alone.

- **Race condition in an asynchronous KYB-verification microservice (least plausible as the dominant cause, not excluded for smaller deltas).** In principle, a refresh/verification event could be recorded before the onboarding transaction commits — for example, if a KYB check fires against a provisional application ID ahead of the final onboarding_date write. This mechanism typically produces discrepancies on the order of seconds to hours, not weeks to months, and does not explain the 22 of 30 cases with deltas ≥ 16 days. It remains a plausible contributor to the four smallest-delta cases (M3051: 4 days; M3085: 7 days; M3037 and M3076: 12 days each), but should not be treated as the primary explanation for the pattern as a whole absent service-log evidence.

10.4 Remediation

- **Database-level integrity constraint.** Add a CHECK constraint enforcing `last_kyb_refresh >= onboarding_date` at the schema level:
- `ALTER TABLE merchants_kyb ADD CONSTRAINT chk_kyb_refresh_after_onboarding CHECK (last_kyb_refresh >= onboarding_date) NOT VALID;`
- The constraint should be added NOT VALID (or equivalent) initially, since the 30 existing violating rows would otherwise block creation; it is validated (VALIDATE CONSTRAINT) only after the batch correction in the item below is complete, so the constraint is never silently bypassed or left unenforced.
- **Write-path audit.** Data Engineering should trace every code path that writes `last_kyb_refresh` — the KYB refresh batch job, any onboarding-service write, and any migration/backfill script — to identify which process is authoritative, confirm its scheduling and idempotency logic, and confirm whether it can execute against merchant records that have not yet completed onboarding. This audit should determine which of the Section 10.3 hypotheses is correct before the correction in the next item is finalized.
- **Batch reprocessing with audit trail — no silent overwrite.** The 30 affected merchant records must not be corrected by direct update. Instead:
- Flag the 30 affected merchants with a `kyb_refresh_status` = “Under Review — Timestamp Correction Pending” marker, suspending reliance on the current `last_kyb_refresh` value for risk-rating refresh-cadence calculations in the interim.
- Re-run the KYB refresh process for the 30 affected merchants to generate a corrected, onboarding-consistent `last_kyb_refresh` value.
- Log every correction in a dedicated audit table capturing `merchant_id`, prior value, corrected value, corrected_by, correction_timestamp, and root-cause reference, retained under the same recordkeeping schedule as the underlying KYB records (Section 6).
- Route the corrections for **M3004** and **M3056** — both tied to open OFAC/UBO case work — through Compliance Officer sign-off before the correction is applied, given the evidentiary sensitivity of those files.

- Add a recurring (daily) data-quality query flagging any new occurrence of `last_kyb_refresh < onboarding_date` and routing it to Data Engineering, so the constraint's NOT VALID state is never allowed to accumulate a new backlog before the next validation pass.